

Dhruv Goyal

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EDUCATION

Indian Institute of Technology, Kharagpur

B.Tech in Civil Engineering

Dec 2021 – Jun 2025

SKILLS

Programming Languages: C, C++, Python, ROS, SQL

Libraries & Frameworks: Numpy, Pandas, Matplotlib, Scikit-Learn, Jupyter, TensorFlow, Keras, SciPy, NLTK, Ultralytics, Librosa, OpenCV, LangChain, LangGraph, CrewAI, FastMCP, SQLAlchemy, Kafka, Selenium, ChromaDb, Pinecone

Tools & Software: MATLAB/Simulink, ROS, GitHub, PostgreSQL, MCP Servers, Docker

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Probability & Statistics

EXPERIENCE

Zolve Innovations Private Limited

AI Engineer

Sept 2025 – Present, Full-time

Multi-Agent Infrastructure & Enterprise AI Systems

- Built a 4-stage NLP-to-SQL pipeline on **Presto/Trino** using RAG over **5K+** query logs, improving SQL accuracy from **68%** to **87%**
- Engineered **AML** & Google Workspace MCP server, Slack forensic reporting, OAuth authentication, deep Google Sheets API integration
- Built LangGraph-based multi-agent framework supporting **20+** configurable **workflows**, tool **routing** & agent state management
- Engineered MCP **client** layer with **stdio/SSE** transports, connection pooling & session recovery, achieving **95%+** workflow reliability
- Built **Kafka**-based workflow trigger system with time-aware scheduling, message parsing, **daemon-thread** execution for agent workflows

AI-Powered User Acquisition Systems

- Created **Reddit** social engagement tool, deploying persona-segmented accounts to drive organic **brand** presence via LLM-based actions
- Built scraping pipeline on **LinkedIn**, **MyVisaJobs**, **Yocket** via API reverse-engineering & Selenium, aggregating **400k+** potential leads

Artificial Intelligence Institute, National University of Singapore

AI Engineer

Jun 2026 – July 2026, Internship

- Build 3-block **CNN** binary classifier over 64×64 **DAPI** tiles, validated via 5-fold **Leave-One-Image-Out CV**, lifting F1 by **27%**
- Deployed model **inference** on **Modal** microservice, bypassing a **150-second** timeout, and optimized the serving pipeline
- Architected pipeline on **HuggingFace** with on-demand display/DZI generation via **OpenSeadragon**, **99.93%** reduction in peak memory

Department of Artificial Intelligence, IIT Kharagpur

Research Intern

Aug 2024 – Dec 2024, Internship

- Developed data generation pipeline via LangChain with **Llama-3.1**, **Phi-3** & one-shot prompting with **regex** for preprocessing data
- Built text analysis system using Named Entity Recognition from **spaCy**, Part-of-Speech tagging for entity frequency distribution
- Visualized data by text clustering using **TF-IDF** & **Word2Vec** embeddings with **KMeans** & PCA dimensionality reduction
- Fine-tuned **RoBERTa** to score data persuasiveness & improved scam detection across highly persuasive synthetic v2 by **65%**

Titan Company Limited

Data Analyst

May 2024 – Jun 2024, Internship

- Performed skewness transformation, **time series decomposition**, & **correlation** analysis, optimizing feature selection & visual insights
- Conducted hypothesis testing with **ADF**, **KPSS** & **ANOVA** tests to identify significant sales drivers, enhancing model accuracy
- Implemented **ARIMA**, **SARIMA**, **LSTM**, **GRU** models for forecasting & addressing patterns, to achieve accuracy of about **83%**
- Incorporated calendar events for holidays/festivals, using **SARIMAX** modeling & pre/post analysis, improving forecasting accuracy **89%**

COMPETITIONS

Trumio.AI Event – Inter IIT Tech 12.0

Gold Medal

Oct 2023 – Dec 2023

- Automated JSON schema generation of project **KPIs** via **OpenAI Function Calling**, enhanced project tracking & contract finalization
- Developed **Resume**, **GitHub** score calculator with Text **Embedding Ada-002**, GPT-3.5-turbo & **FAISS**, enhancing selection accuracy
- Developed **LLM-driven** roadmap system that generates personalized learning paths & live adaptive tests to improve user and team skills

The American Express Campus Challenge 2024

Qualified Round 2

Jun 2024 – Jul 2024

- Engineered over **70+** features from key variables, applied **ANOVA** & **Chi-Square** tests for correlations, enhancing model precision
- Utilized **XGBoost** & **SHAP** for comprehensive feature importance evaluation, achieved over **65%** accuracy in predicting winners

AWARDS & ACHIEVEMENTS

- Recipient of Silver Medal in Product Development in General Championship of Technology, 2023, designed Vaccine Transport Box
- Competed in the RoboSumo event organized by the Technology Robotix Society, securing an impressive 5th position among Institute
- Actively participated as a member of the Gold-Winning Cricket Team of Rajendra Prasad Hall in the General Championship, 2023.